Bell Nozzle Rocket Engine Calculations

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Contents

Nomenclature

C_f	Vacuum thrust coefficient
$\dot{m}$	Mass flow rate
M	Molar mass
Ma	Mach number
$mol\%$	Mole percentage
P	Pressure
R	Gas constant
T	Temperature
v	Velocity
ϵ	Nozzle area ratio
γ	Ratio of specific heats
ρ	Density

Subscripts

$_a$	Ambient
$_c$	Chamber
$_e$	Exit
$_s$	Stagnation
$_t$	Throat

Introduction

Documentation of calculations for a bell nozzle rocket engine

1 Propellant

1.1 Propellant Stoichiometry of Hydrocarbon-Oxygen Fuel Mixes [1]

$$C_nH_{2n+2} + \frac{3n+1}{2}O_2 \rightarrow (n+1)\,H_2O + nCO_2 + energy \tag{1.1}$$

2 Exhaust Gasses

2.1 Ratio of Specific Heats of Multi-Compound Gas Mixtures [2]

$$\frac{1}{\gamma - 1} = \sum \frac{mol\%}{\gamma_i - 1} \tag{2.1}$$

2.2 Stagnation Pressure of Exhaust Gasses [3]

$$P_s = P_1 \left(1 + \left(\frac{\gamma - 1}{2}\right) Ma_1^2\right)^{\frac{\gamma}{\gamma - 1}} \tag{2.2}$$

2.3 Stagnation Temperature of Exhaust Gasses [4]

$$T_s = T_c \left(1 + \left(\frac{\gamma - 1}{2}\right) Ma_1^2\right) \tag{2.3}$$

3 Combustion Chamber Geometry

4 Throat Geometry

5 Nozzle Geometry

[2]

Bibliography

References

[1] J. Hawley, "The stoichiometry of burning hydrocarbon fuels." https://jimhawley.ca/downloads/Diesel/The_stoichiometry_of_burning_hydrocarbon_fuels.pdf, 2014.

[2] O. Lanzi, "How do you calculate the heat capacity ratio for a multi-compound gas?." https://chemistry.stackexchange.com/a/166796, 2022.

[3] H. Vasishta, "Stagnation pressure." https://www.youtube.com/watch?v=t98s1A7fvm0, 2016.

[4] H. Vasishta, "Stagnation temperature." https://www.youtube.com/watch?v=Le5dgW4lMWs, 2018.